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Full paper

Highly dispersed porous polydimethylsiloxane for boosting power-generating performance of triboelectric nanogenerators

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ABSTRACT

One of the common approaches in improving the output performance of triboelectric nanogenerators (TENGs) is to increase triboelectric charge density by enlarging the surface area of triboelectric materials. Herein, highly porous polydimethylsiloxane (PDMS) introduced by nanograss silicon mold is developed to improve the output performance of PDMS-based TENGs in contact-separation mode. With an area of 100 cm^2, the power generating performance of porous PDMS-based TENG (S1) is greatly enhanced and the maximum voltage of 3200 V and current of 94 μA are obtained. It also delivers a maximum power density of 0.58 W m^{-2} at a load resistance of 10 MΩ, offering almost three times power improvement compared with the TENGs with micro-pillar (S2) and flat (S3) PDMS surfaces. The boosting power-generating performance is mainly attributed to an increase surface roughness and additional triboelectric charges induced on the surfaces of the inner pores. Additionally, the S1 TENG is successfully used as a power source to directly drive 400 series-connected light emitting diodes (LEDs) to high brightness. This developed TENG also provides enough energy for 10 μF capacitor to power a wireless transmitter within 10 s. This work presents a new porous patterned PDMS-based TENG, which not only can be readily fabricated but also considerably delivers the high output performance toward the realization of self-powered and sustainable electronic devices.

1. Introduction

The continuing development of integrated circuits for several decades has been the main driving force for miniaturized electronics that have resulted in not only an increase in computing power but also a decrease in energy consumption. Miniaturized electronics operating with ultralow power consumption have also made the self-powered systems possible by scavenging from their surrounding mechanical energies [1]. Among several mechanical energy harvesting technologies, triboelectric nanogenerator (TENG) based on the contact electrification and electrostatic effects [2,3] emerges as a promising harvesting technique to convert mechanical energy into electricity [4] due to its extremely high output voltage, simple design and fabrication, light weight, flexibility and low cost [5–8]. The TENGs have successfully transformed various types of mechanical movements into electricity such as touching [9,10], impact [11,12], linear sliding [13,14], rotation [15,16] and vibration [17,18]. However, further improvement in the output performance is essential to speed up the realization of self-powered electronics for practical applications.

Enhancing triboelectric charge density via carefully selecting the two triboelectric materials with a large electronegativity difference from the triboelectric series or increasing the effective contact area of the friction layers are regarded as effective approaches to boost the TENG performance [19–22]. Generally, polymer and metal are chosen as triboelectric materials in high performance TENGs because they are at the opposite end of the triboelectric series. Compared to other polymers, polydimethylsiloxane (PDMS) has several advantages such as the excellent electron accepting property, flexibility and biocompatibility, making it a potential candidate for wearable applications. A solid PDMS also possesses a hydrophobic surface which helps maintain the electrical

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output of PDMS-based TENGs under high humid conditions.

Another approach to improve the TENG performance is by enlarging the friction area of triboelectric materials. Several surface patterning techniques have been employed to serve this purpose such as the use of silicon mold for pattern transfer [23,24], plasma etching [25,26], chemical treatment [27,28]. The silicon mold is usually preferred because hundreds of replicas of patterned PDMS can be produced from one single mold, therefore making it environmental-friendly and easy for mass production and fabrication. The enhanced output of TENGs with micro- and nano-patterned PDMS has also been reported in the previous work [29,30].

In this work, a simple pattern transfer technique was employed to create highly porous PDMS structure by using a silicon mold with a nanograss pattern which is a forest of silicon nanometer-sized spikes created with reactive ion etching [31]. This nanograss pattern was successfully used as a secondary effect on increasing the roughness of V-shape groove and pyramid PDMS microstructures [29]. Herein, the nanograss-patterned PDMS and Al layers will be employed to construct TENGs in a vertical contact-separation mode because of their distinctly different triboelectric properties [32,33]. Flat and micro-pillar patterned PDMS-based TENGs were also fabricated for comparison purpose. The TENGs containing highly porous PDMS film generates higher voltage and current outputs (~3200 V and ~100 μA) than those from micro-pillar patterned (~1800 V, ~70 μA) and flat films (~1500 V, ~60 μA). The maximum power density delivered from porous PDMS surface is three times higher than that from a micro-pillar and flat surfaces. This generated electricity can directly light up hundreds of green commercial LEDs and power the wireless remote switch within 10 s demonstrating its potential in battery-free application. More importantly, the output performance is maintained even after 20,000 cycles, highlighting the excellent stability and mechanical durability of the TENGs developed in this study.

2. Experimental section

2.1. Fabrication of silicon molds

Two silicon molds with different micro-structures, which are deep circular holes and nanograss arrays, were designed and fabricated in this study. For the mold with circular hole arrays, a silicon wafer was first cleaned using the standard cleaning process, then silicon dioxide (SiO_2) was thermally grown on the wafer and subsequently patterned with the array of circles using photolithography process. The patterned silicon wafer was then dry etched using reactive ion etching in order to obtain anisotropic etch profile (deep circular hole). For the silicon wafer with nanograss arrays, another cleaned silicon wafer was dry etched using the process similar to Zhang et al. [34]. This etching process can create nanograss patterns with ultralow-surface-energy silicon substrate, therefore preventing the bonding between the PDMS and the mold during curing process. The silicon wafers with circular hole and nanograss patterns were later used to create micro-pillar and highly porous PDMS surfaces, respectively.

2.2. Fabrication of PDMS-based triboelectric nanogenerator

The fabrication process of patterned PDMS surface is schematically illustrated in Fig. 1. Once silicon molds were successfully fabricated, they were treated with hexamethyldisilazane (HMDS, J.T.Baker) by gas phase silanization for 1 h to avoid the adhesion between PDMS and the molds. Meanwhile, PDMS elastomer and cross-linker (Sylgard 184, Tow Corning) were mixed in a 10:1 wt ratio and then degassed under vacuum. The degassed PDMS mixture was subsequently casted on the silicon molds and spin-coated at 300 rpm for 60 s to obtain the PDMS film with the thickness of ~130 μm. After curing at 90 °C for 15 min, the uniform PDMS thin film was peeled off from the silicon masters.

The TENGs with three different PDMS surfaces which are porous (S1), micro-pillar (S2) and flat (S3) patterns were constructed and compared for their energy harvesting performance. To fabricate PDMS-based TENGs, the PDMS was laminated on a piece of clean ITO-coated PET film (Sigma Aldrich) to form a bottom electrode which was then supported by expanded polyethylene (EPE) foam. Aluminum foil as a top electrode was supported by acrylic sheet and EPE foam. All fabricated TENGs are assembled in contact-separation mode with an area of $10 \times 10\ cm^2$. The top and bottom structures of TENG were attached to the top and bottom plates of the pushing tester of which the top part can be moved up or down to achieve desired distances. The conducting wire was then connected to the top and bottom electrodes with Kapton tape and alligator clips, respectively, for electrical characterization.

2.3. Electrical and material characterizations

For the electrical measurement of TENG, an external force was applied vertically through the pushing tester to create periodic contact and separation between the top and bottom structure of TENG. This periodic motion generated voltage and current outputs which were measured by digital oscilloscope (Agilent 2002A) and Keysight B2901A, respectively. The surface morphology of patterned PDMS films was examined by scanning electron microscopy (SEM) on JEOL JSM-6335 F.

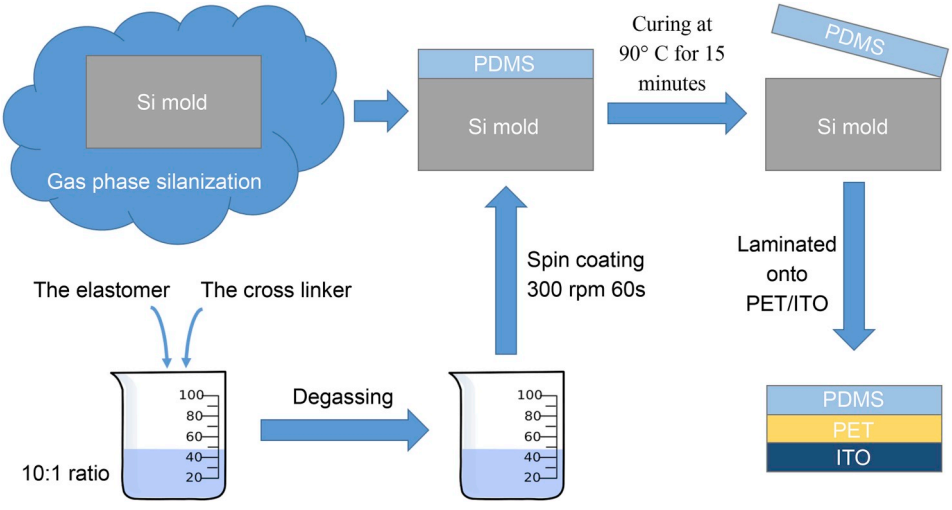


Fig. 1. Schematic fabrication process of patterned PDMS and the bottom electrode for TENG.

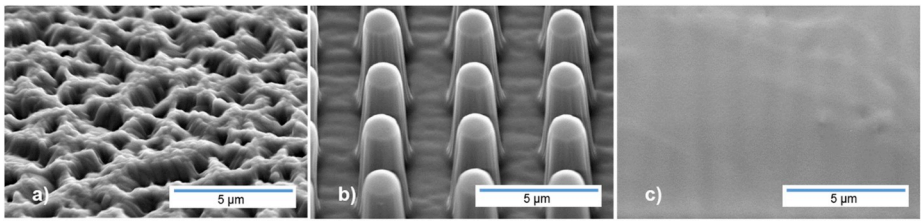


Fig. 2. SEM images of the patterned PDMS a) with porous morphologies (S1) created by nanograss silicon mold and b) with pillar structures (S2) created by circular silicon template compared with that of c) flat PDMS.

3. Results and discussion

The SEM images of S1 (Fig. 2a) and S2 (Fig. 2b) reveal porous morphologies and pillar features created by nanograss and circular hole silicon molds, respectively. Both patterns show high uniformity across PDMS surface. For comparison purpose, the SEM image of flat PDMS (S3) is also shown in Fig. 2c. The operation of TENG in contact-separation mode is realized by applying a periodic vertical force on the whole area of the TENG device. Since there is no spacer, the complete contact between two triboelectric materials is expected. The working principle of the TENG under periodic contact and separation is demonstrated in Fig. 3.

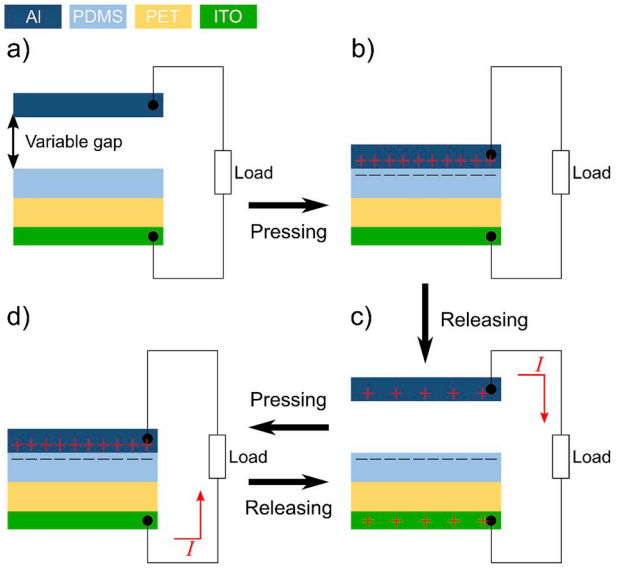


Fig. 3. Illustration of the operating principle of patterned PDMS-based TENGs. An alternating current is generated under periodic contact and separation.

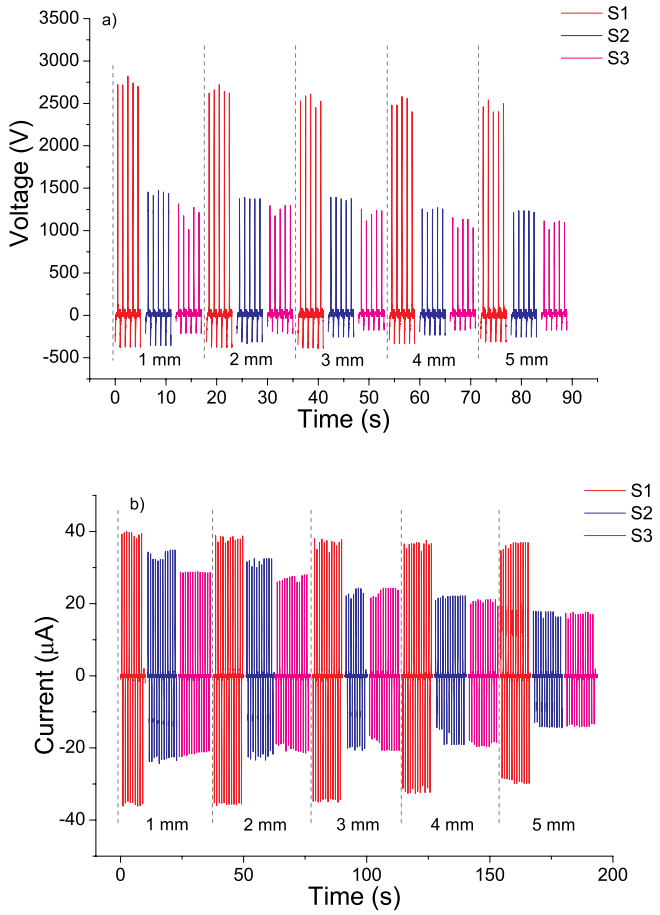


Fig. 5. a) Output voltages and b) currents measured at 1 Hz frequency from the TENGs with three different PDMS surface morphologies as a function of spacing between aluminum and PDMS.

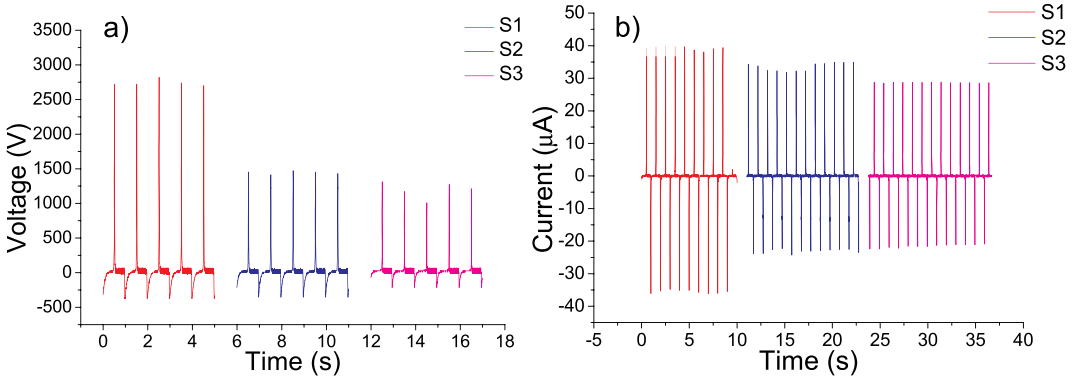


Fig. 4. Output performance of the TENGs with 1 mm gap and 1 Hz frequency. a) Output voltages and b) currents of TENGs with the porous (S1), pillar (S2) and flat (S3) surfaces.

Before the two triboelectric layers are in contact, there is no charge transfer between the two materials, thus a zero-electric potential. Once the aluminum layer is in contact with PDMS, the PDMS attracts electrons from aluminum due to differences in electron affinity. This charge transfer process results in positive and negative triboelectric charges on the surface of aluminum and PDMS, respectively. At this stage, there is no current generated yet because the TENG remains in electrostatic equilibrium state. When the applied force is released, a potential difference is created between top (Al) and bottom electrode (ITO) due to the separation of opposite surface charges. As there are positive triboelectric charges on aluminum, electrons from ITO electrode are therefore attracted to flow through an external circuit to aluminum electrode, generating an electrical signal. Subsequently, when the force is again applied to the TENG, electrons flow back from aluminum electrode to ITO electrode through the external circuit. An opposite electric signal is then observed. Therefore, the TENG has an ability to generate an alternating current (AC) through periodic contact and separation motion created by an external force.

The fabricated TENGs with three different PDMS surface patterns (S1, S2 and S3) were electrically characterized with controlled amplitude of applied vertical force. With 1 mm distance between aluminum and PDMS, the output voltages and currents of the three fabricated TENGs at 1 Hz frequency (Fig. 4) show that the S1 and S2 TENGs with patterned PDMS produce higher voltage (Fig. 4a) and current (Fig. 4b) than the S3 with flat PDMS surface due to their roughened surfaces and an improved contact electrification area as a consequence. The S1 TENG with highly porous PDMS film delivers the best performance with the maximum peak-to-peak voltage of ~3200 V and the maximum peak-to-peak current of ~76 μA. Such high output performance can be ascribed to the unique morphology of porous PDMS which provides high surface roughness, thus enhancing the contact electrification area. It is possible that the additional charges on the inner pore surfaces generated as a result of electrostatic effects associated with the restoration of pore shape after contact-separation motion may also contribute to the superior performance of the porous PDMS-based TENG [35]. In comparison with that of the porous PDMS, a lower electrical output is found for the micro-pillar PDMS. This low performance is ascribed to the susceptibility to surface and external forces (i.e., adhesive and capillary forces, and compressive forces) of this high-aspect-ratio pillar PDMS, leading to pattern collapse and structural deformation [36–38]. As seen from Fig. S1, the ground and lateral collapses of the pillar PDMS can be clearly observed after operating cycle.

The effect of gap between aluminum and PDMS on the electrical performance was also studied. The output voltages and currents of the

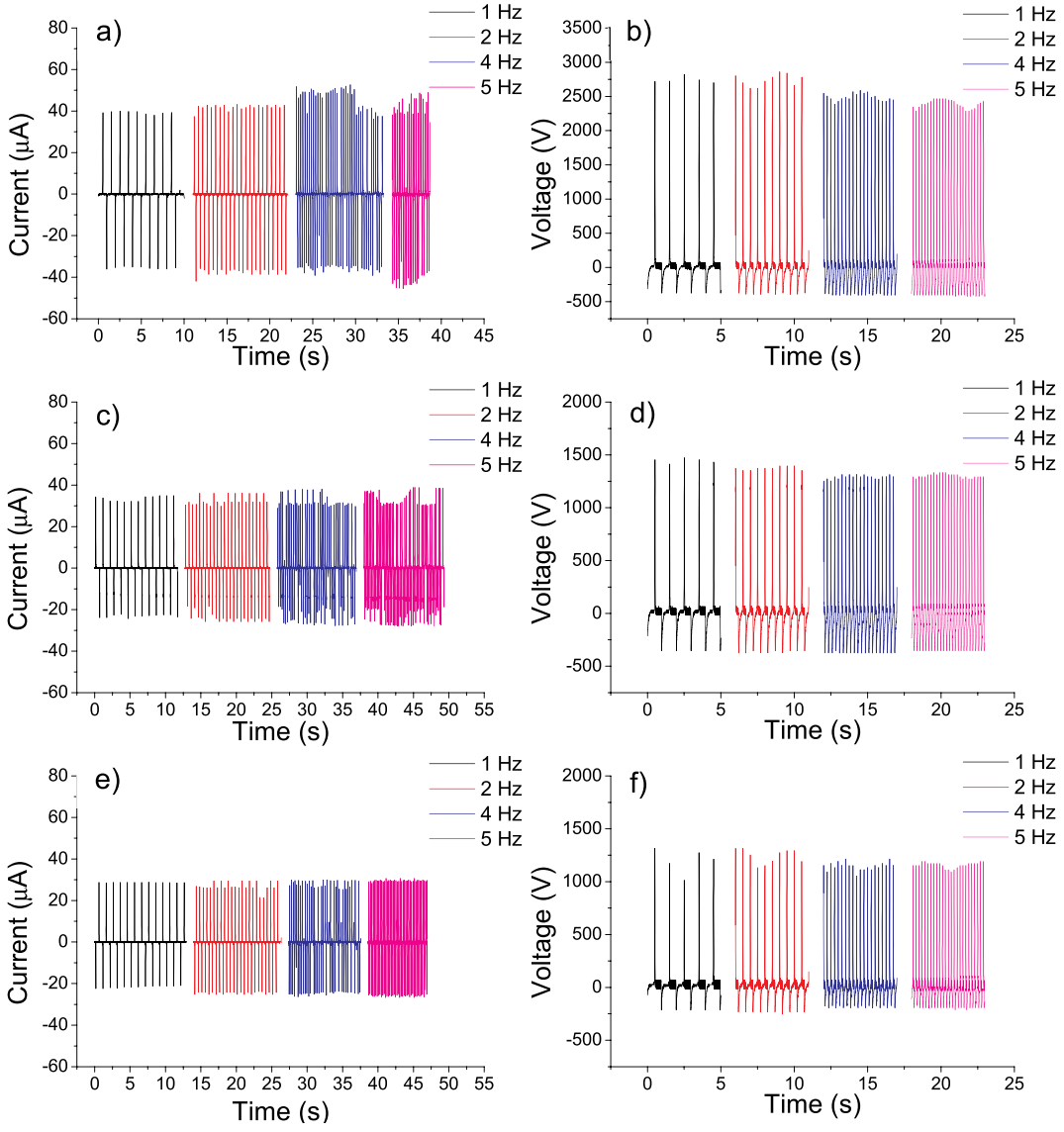


Fig. 6. The output currents and voltages of S1 (a–b), S2 (c–d) and S3 (e–f) TENGs as a function of frequency with the 1 mm gap.

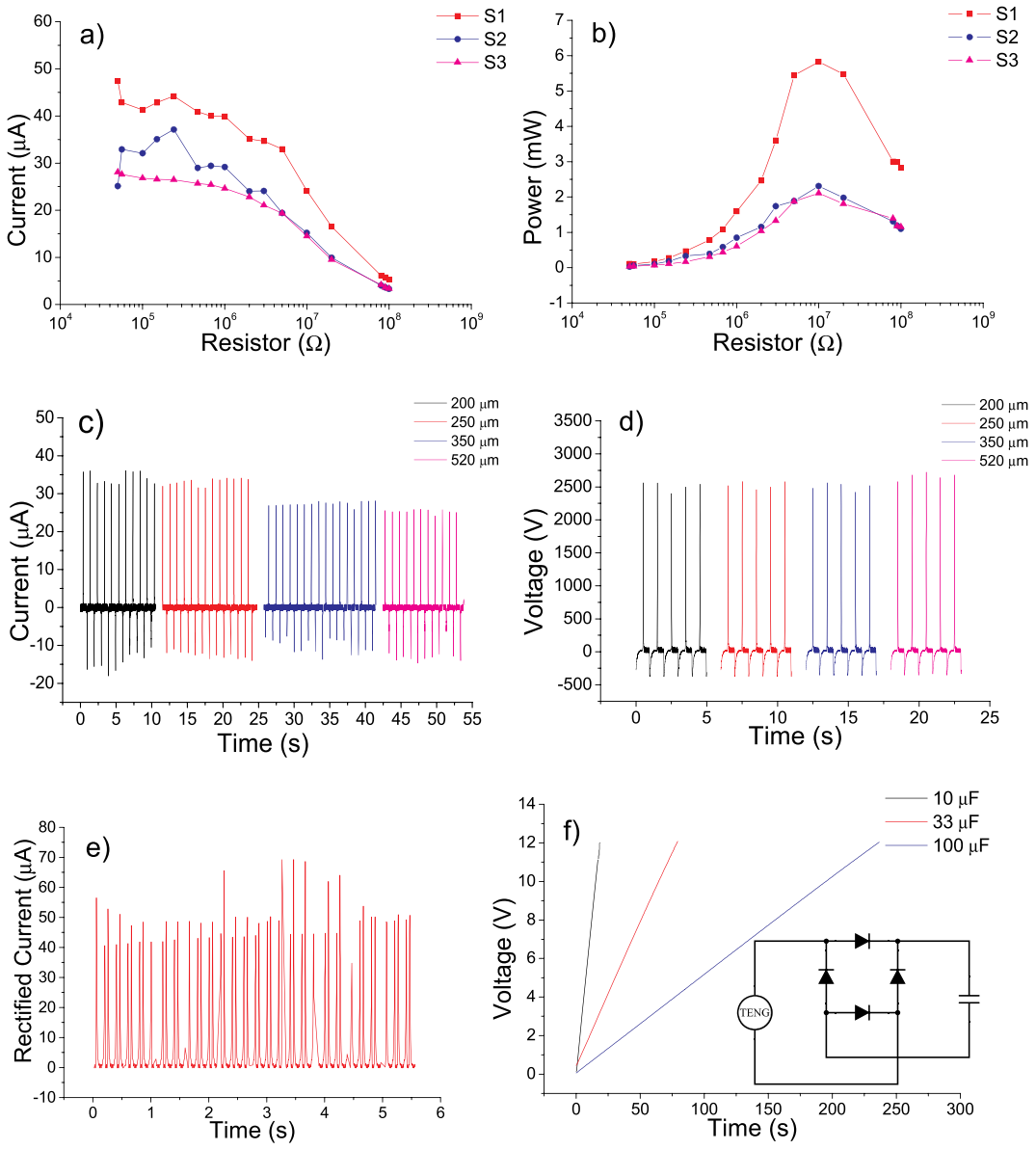


Fig. 7. a) The output currents and b) the output powers of S1, S2 and S3 TENGs as a function of external load resistance. c) Output current and d) voltage of S1 TENGs fabricated from porous PDMS with different thicknesses at a frequency of 1 Hz and 1 mm gap. e) Rectified current and f) the capacitor charging behaviors of the S1 TENG.

TENGs with different separation distances at 1 Hz frequency are displayed in Fig. 5. The dependence of the TENG output performance on the gap is clearly observed on all samples. The output voltage and current are gradually decreased as the distance increases because, with an increasing space, a longer time is required for the two triboelectric materials to make contact, therefore more surface charges are decayed leading to the reduction in both voltage and current. Similar phenomena were also discovered from Hou et al. [39] and He et al. [40]. Theoretical study [41] has previously shown that the average output power increased as the gap between two triboelectric materials reduced and the maximum power was obtained at 1 mm gap which is consistent with our experimental results. Therefore, the 1 mm spacing is used throughout in this study.

To determine the optimum operating conditions, the output dependence of TENGs on frequency has also been investigated. The pushing frequency employed are 1, 2, 4 and 5 Hz while the gap is fixed at 1 mm with the identical applied force. It can be clearly seen from Fig. 6a, Fig. 6c and e that the amount of current increase for S1 is ~20 μA (~76 μA for 1 Hz to ~94 μA for 5 Hz) which is much higher than that of S2 (~7 μA) and S3 (~6 μA). The current increase at high frequency is attributed to higher flow rate of triboelectric charges. Although an amount of triboelectric charges are constant because of the fixed materials and gap, the higher flow rate of charges leads to the increased current [23]. As shown in Fig. 6b, d and f, the peak-to-peak voltage of all TENGs changes slightly at different frequencies because the charge density and separation of materials, which influence the output voltage, are not significantly affected by the pressing frequency. Similar behaviour is also observed from the work reported by Wang et al. and Chen et al. [23,42].

The TENGs with 1 mm gap were connected to resistor ranging from 5 kΩ to 100 MΩ to further investigate the dependence of the output on a resistive load. From Fig. 7a, the output current decreases as the resistance increases and all devices display a similar trend. Within the 5 kΩ–1 MΩ, the output current slightly reduces because the charge transfer process is relatively similar to the short-circuit condition. As the load resistance rises to the range of 2–20 MΩ, the charge transfer between

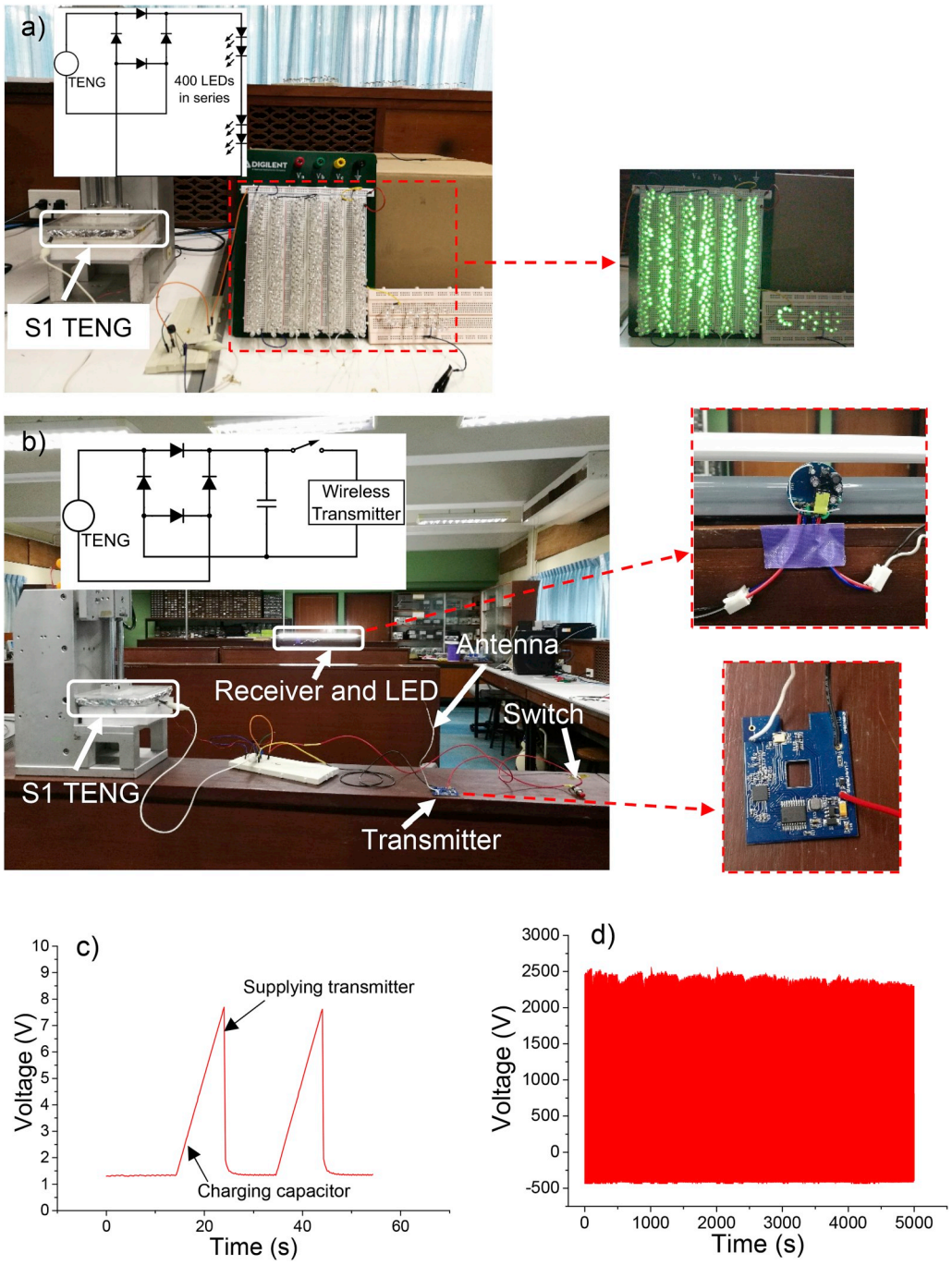


Fig. 8. a) The schematic and photograph of 400 series-connected LEDs directly driven by the S1 TENG. b) The schematic and photograph of the S1 TENG charging 10 μF capacitor, which supplies power to operate a wireless transmitter. c) The capacitor voltage characteristics during the operation of wireless transmitter. d) Stability test of the S1 TENG over 20,000 cycles at 5 Hz.

two electrodes is significantly diminished resulting in a drastic drop of current. The output current is close to zero as the resistor approaches the open-circuit condition (>20 MΩ). The observed output current as a function of resistance is in agreement with theoretical study reported by Niu et al. [43]. The corresponding output power as a function of external load resistance is calculated and plotted in Fig. 7b. All three TENGs deliver their maximum output power at the load resistance of 10 MΩ. The maximum power density of 0.58 W m^{-2} for the S1 is almost three times higher than that of S2 (0.23 W m^{-2}) and S3 (0.21 W m^{-2}), ensuring the superior performance of the TENG with porous-structured PDMS. The power density of our S1 TENG is greater than that reported in Ref. [44] (0.21 W m^{-2}) where the femtosecond laser direct writing technique was employed to create surface patterns on both PDMS and Cu films. Compared with that reported previously [21,25,26], the main advantage of porous PDMS developed in this work is probably its markedly high surface roughness which results in enlarged contact electrification area between two triboelectric materials, therefore an improved electrical output. This is supported by the AFM investigation in Fig. S2. The average surface roughness (R_a) of our porous PDMS is 199 nm, which is an order of magnitude higher than that in other reports (75.6, 29.7 and 27.4 nm in Ref. [21,25, 26] respectively).

The effect of PDMS film thickness on the output performance of porous PDMS-based TENG was also studied as shown in Fig. 7c and d. As increasing the PDMS thickness from *ca.* 130 μm to 200, 250, 350 and

520 μm, the current is gradually decreased, however the output voltage remains nearly the same. Our results correspond well with the theoretical model proposed in Ref. [43] where the increased effective dielectric thickness reduces the output current, but not the voltage.

The ability of the S1 TENG with 1 mm gap in charging a capacitor is also investigated. After the AC current of the TENG was converted into direct current (DC) by a bridge rectifier circuit (Fig. 7e), the harvested energy is then used to charge different capacitors. As seen from Figs. 7f, 10, 33 and 100 μF capacitors can be charged at a pushing frequency of 5 Hz–12 V in 18, 78 and 234 s, respectively.

To comprehensively evaluate the performance of the developed TENG in practical applications, the S1 TENG was used as a power source to light up commercial light emitting diodes (LEDs) and power a wireless transmitter. As shown in Fig. 8a and Video S1 in the supplementary information, the S1 TENG is capable of directly driving 400 LEDs connected in series to high brightness. Furthermore, the generated energy can successfully be used to supply a wireless transmitter. By charging a 10 μF capacitor to 7 V at 5 Hz in 10 s, the transmitter is powered up and a signal is simultaneously sent to the receiver, thus turning on the LED lights (Fig. 8b and Video S2). The variation of capacitor voltages for two consecutive transmitting processes are shown in Fig. 8c. The time required for charging the capacitor is in the similar range to that reported by Ling et al. [45]. However, as their device structure is more complicated than ours, the results here thus underline our simple designed porous PDMS-based TENG with boosting performance. The stability of the S1 TENG is also examined as shown in Fig. 8d. Less than 10% decrease in the output voltage is observed, after more than 20,000 cycles of a continuous operation at 5 Hz.

Supplementary video related to this article can be found at https://doi.org/10.1016/j.nanoen.2019.104214

4. Conclusion

The highly porous PDMS-based TENG device with remarkable performance has successfully been fabricated using nanograss silicon mold. Electrical characterization shows that the TENG with porous PDMS surface produces the highest voltage, current and power density of 3200 V, 94 μA and 0.58 W m^{-2}, respectively, compared with the TENGs with pillar and flat PDMS surfaces. The enhanced performance is attributed to increased effective contact electrification area and additional triboelectric charges induced on the surface of the inner pores. More importantly, this porous PDMS-based TENG is successfully used as a power supply for 400 series-connected LEDs and a wireless transmitter, opening up an alternative power source for self-powered and sustainable electronic systems. The present work introduces a new porous patterned PDMS film for the improvement of TENG output performance and intensifies the benefit of highly porous surface which would further encourage the practical applications of PDMS-based TENG devices.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nanoen.2019.104214.

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